Network Science Final Project Report

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CSCI 399

PART 0 & 1:
Basic Information

This project contains a network of nodes and edges representing player transitions between teams in Major League Baseball (MLB), ranging from 1994 to 2025. This dataset was compiled from several websites, and I combined the data into two CSV files: one for all trade data and one for winning percentages. The trade information was collected from "thebaseballcube.com", which contained CSV data for each year in the MLB, and "baseball-reference.com", where I used a script made by AI to scrape the website to collect players' WAR post-trade to obtain the "value" they provided after the trade. The site was also used to obtain each team's winning percentage over the past 30 years. I then organized this data into two CSV files for this project. This network is a weighted, directed multigraph with 30 nodes representing each MLB team and 9269 edges representing players traded from team to team, with direction, labeled with the player's name, and weighted by the stat WAR (Wins above replacement). WAR is a single number that estimates how many more wins a player gives their team compared to a "replacement-level" player. Replacement is a bench player or minor leaguer. 0 is the baseline, which explains why there are so many in the data set. WAR is a combination of many things:

For hitters:
- Batting (hits, power, getting on base)
- Baserunning (stealing bases, taking extra bases)
- Defense (fielding ability)
- Position difficulty (shortstop > first base)

For pitchers:
- Preventing runs (most important)
- Innings pitched (durability)
- Context-adjusted performance (Errors by infield, pouring rain)

Positive WAR means they are helping their team win compared to a baseline player
Negative WAR means they are costing their team wins

For these reasons, I feel WAR properly represents the value the team received from the player, serving as a good benchmark for the trade's value and for determining whether a team "won" the trade. Along with WAR, I have also collected each team's winning percentage and number of championships over the last 31 years, allowing me to examine any correlation between WAR acquired through trades and overall team success.

As a baseball fan, I am very interested in this data set. I want to see if there is any correlation between the number of players traded away and winning % over the years. I would also like to visualize which team is "better" at trading solely based on the total war each team has gained and lost over the years through trades, and see whether winning trades correspond to an increase in winning percentage.

With this data, I ran into some hiccups. One criticism I had was the number of zero values for players' WAR stat, which I realized the scraper I used couldn't collect data for players with abbreviated names (e.g., J.A. Happ). To fix this, I had to manually adjust the players' WAR. The other zeros can be attributed to trading minor leaguers who never made it to the big leagues (most common), to players who didn't get ample playing time, and to those who were truly baseline or replacement level. Then, to collect each team's winning percentage over the years, I was unfortunately unable to have AI generate me code to scrape the information, so I went in manually to collect each season's data, and with a little help from Google Sheets commands, I got it all formatted properly.

Visualizations:

To visualize this data, I created two networks: one containing all trades over the 31 years and another containing all trades for a single year. Pictured below are the two networks. Figure 1 contains every trade, and Figure 2 shows every trade in 1994. The network for the single year is adjustable each year, but for simplicity, I picked the first year.

Figure 1:

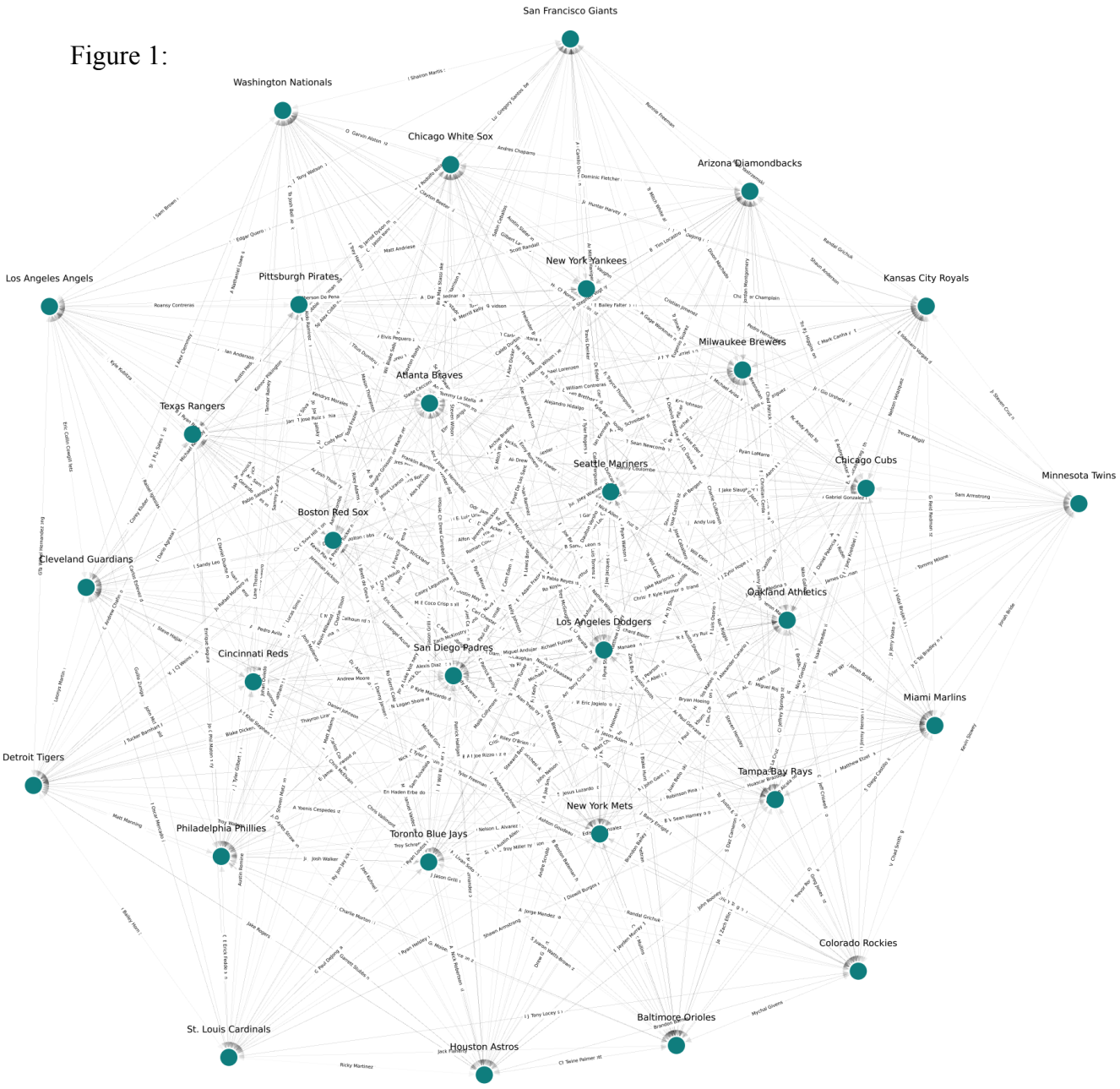
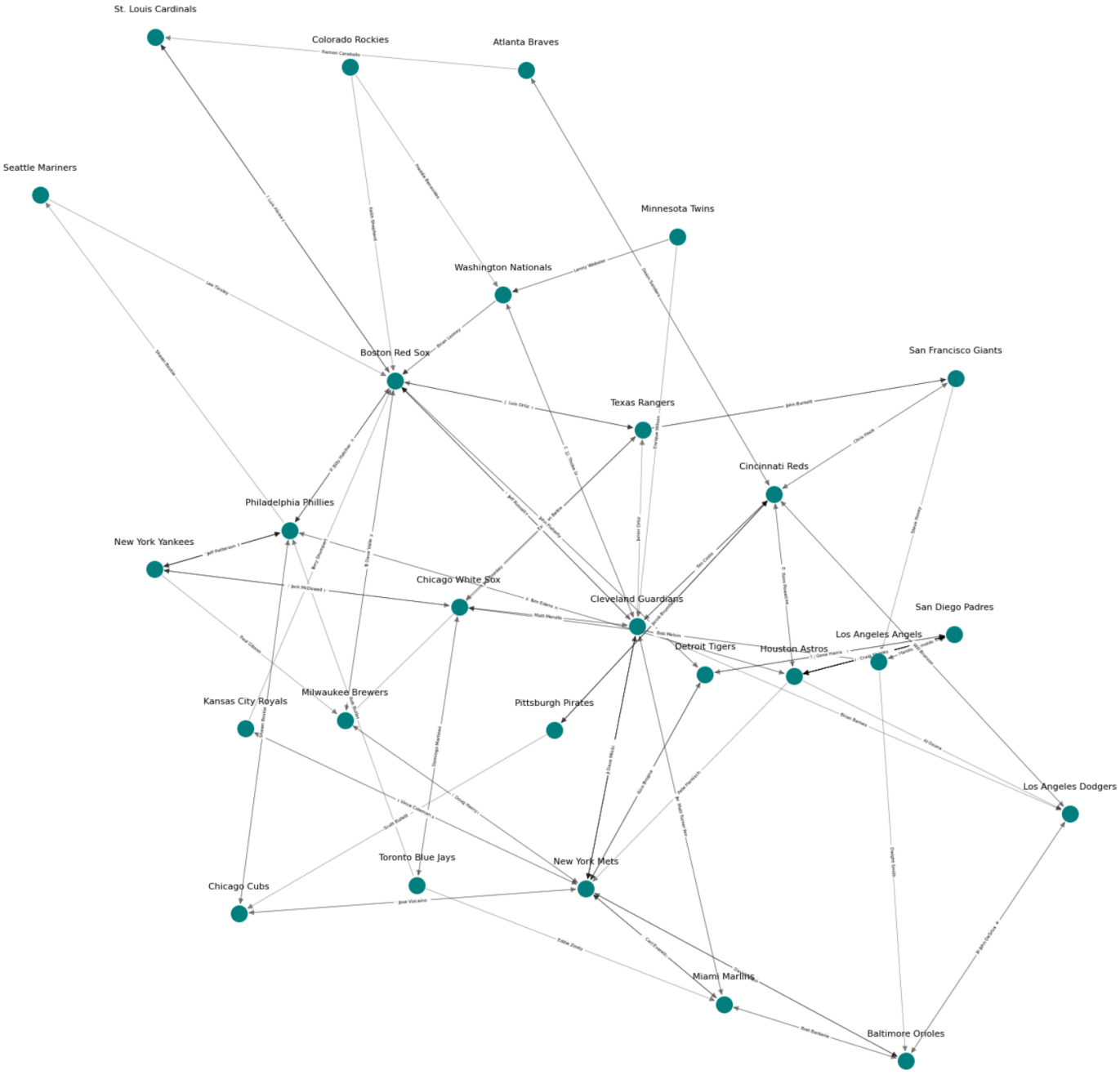


Figure 2:


St. Louis Cardinals
Colorado Rockies
Atlanta Braves
Seattle Mariners
Minnesota Twins
Washington Nationals
Boston Red Sox
San Francisco Giants
Texas Rangers
Cincinnati Reds
Philadelphia Phillies
New York Yankees
Chicago White Sox
Cleveland Guardians
San Diego Padres
Los Angeles Angels
Detroit Tigers
Houston Astros
Kansas City Royals
Milwaukee Brewers
Pittsburgh Pirates
Los Angeles Dodgers
Toronto Blue Jays
New York Mets
Chicago Cubs
Miami Marlins
Baltimore Orioles

I then used this data to compute the total number of players traded for and away, the WAR each player brought to each team in the trade, and the net WAR for each team. Below are the stats for all trades from 1994 to 2025:

Team	Num Players Traded For	Num Players Traded Away	War Received	War Given Away	Total WAR From Trades
San Diego Padres	399	437	461.46	399.72	61.74
Oakland Athletics	373	338	448.46	460.43	-11.97
Seattle Mariners	367	399	277.74	501.19	-223.45
Boston Red Sox	363	367	382.26	424.12	-41.86
Los Angeles Dodgers	359	399	332.11	230.74	101.37
Miami Marlins	350	316	328.04	607	-278.96
Tampa Bay Rays	350	316	328.93	363.72	-34.79
Baltimore Orioles	342	244	226.26	138.47	87.79
New York Mets	341	363	299.22	268.11	31.11
New York Yankees	339	366	413.06	316.15	96.91
Pittsburgh Pirates	331	330	217.08	296.28	-79.2
Atlanta Braves	330	346	317.63	280.89	36.74
Milwaukee Brewers	329	308	334.66	267.79	66.87
Texas Rangers	327	338	312.95	391.9	-78.95
Toronto Blue Jays	314	347	358.65	349.24	9.41
Cincinnati Reds	313	301	277.71	250.25	27.46
Chicago Cubs	305	343	309.94	289.62	20.32
Chicago White Sox	301	296	376.63	256.54	120.09
Philadelphia Phillies	299	300	254.15	282.82	-28.67
Arizona Diamondbacks	294	325	288.25	416.49	-128.24
Cleveland Guardians	292	323	484.09	408.58	75.51
Kansas City Royals	290	290	185.89	187.55	-1.66
Detroit Tigers	272	241	336.48	261.03	75.45
Washington Nationals	262	252	199.63	327.49	-127.86
Colorado Rockies	258	242	168.01	193.67	-25.66

Houston Astros	257	263	299.17	241.82	57.35
Los Angeles Angels	253	224	142.3	176.23	-33.93
San Francisco Giants	241	235	251.79	169.71	82.08
St. Louis Cardinals	209	213	369.97	284.06	85.91
Minnesota Twins	208	206	238.5	179.41	59.09

I have also found mean, median, and standard deviation information for each network. The data below is for every trade since 1994. This data shows that trades are generally balanced across the league, but not at the team level. Most teams gain a moderate amount of WAR from trades, while a small number lose large amounts, creating a highly skewed distribution. Overall, this suggests trades tend to benefit the majority of teams slightly, with a few teams absorbing significant losses that offset those gains.

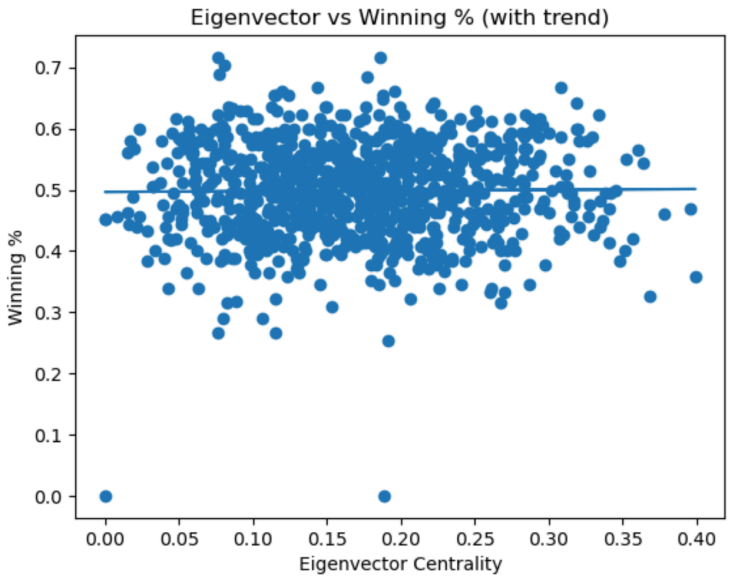
	Players traded away	Players traded for	WAR given away	WAR received	Netl WAR
Average	308.93	308.93	307.37	307.37	0
Median	313.5	313.5	283.44	311.45	23.89
Standard Deviation	47.24	57.93	106.38	83.05	93.93

PART 2:
Centralities and Community Detection

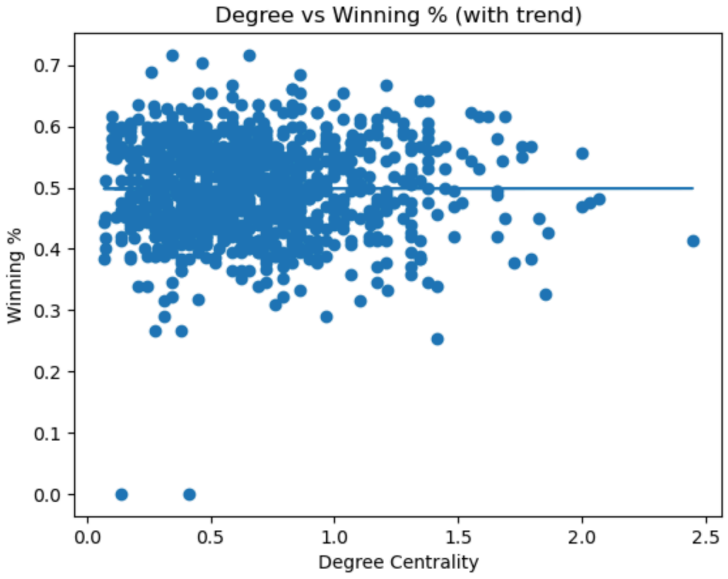
For this section of the project, I wrote a code cell that cycles through each team each year, calculates their Degree, Betweenness, Eigenvector, Katz, and Closeness centralities, and formats the results into a CSV (mlb_centrality_data.csv) file. Degree represents the number of trade relationships a team has. It is calculated by taking the number of neighbors a node has divided by the total number of nodes. Closeness is the average shortest path between two nodes, calculated as the number of nodes divided by the shortest path between them. Betweenness means how often a node lies on the shortest paths between other nodes, calculated by the number of shortest paths from two nodes over the number of nodes passing through the main node. Eigenvector assigns importance to nodes based on their connections to other important nodes: each node receives a score equal to the sum of its neighbors' scores, then the scores are normalized. Katz's is similar to eigenvector but accounts for all paths; it also gives nodes scores, but with a constant to ensure all nodes have a score. Degree, betweenness, and closeness centralities were computed using weighted edges, where weights represent WAR values, while

eigenvector and Katz centralities were computed using unweighted edges due to the use of the adjacency matrix. Then, I used that data to make a scatter plot to visualize the overall centralities, comparing them with that team's winning % to see if there is a trend. Each point in these plots represents a team in a given year, meaning there are multiple data points per team across seasons rather than a single aggregate value per team. The following plots are shown below.

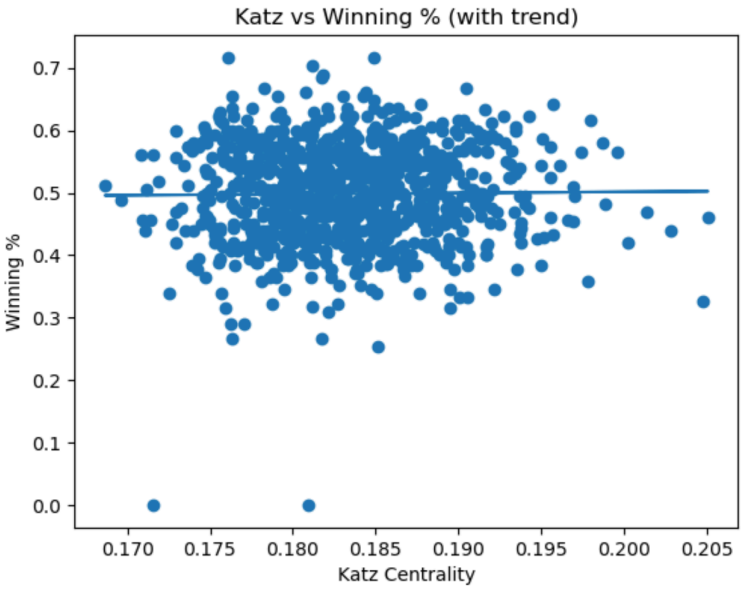
Graphs:



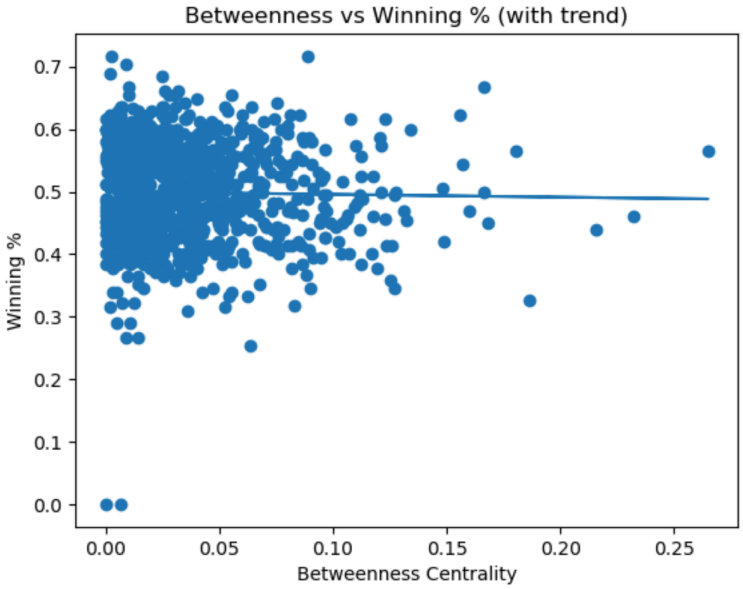
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correlation 0.0009248820362684486

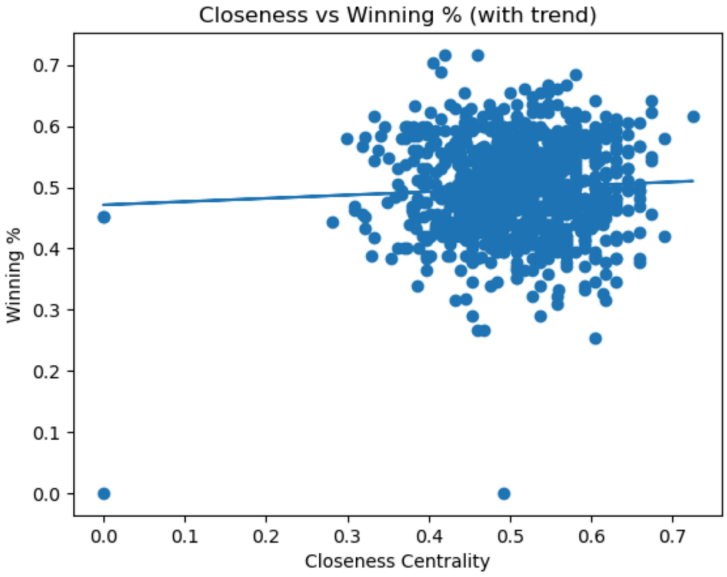


correlation 0.012856417491817575



correlation -0.019213662865509358

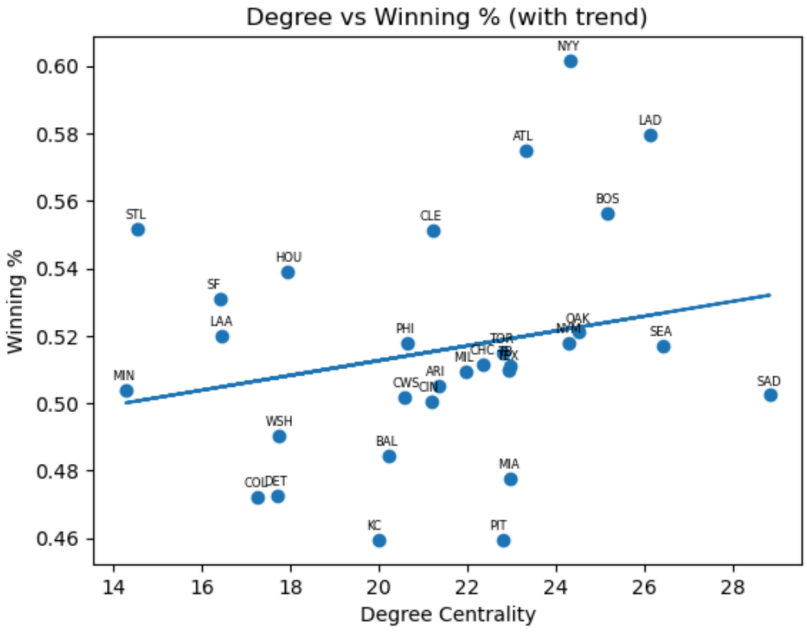
As you can see from the trendline and correlation, there doesn't seem to be any correlation between centralities and winning percentage. So from 1994–2025 common centrality measures exhibit little to no correlation with team winning percentage.



correlation 0.05419129826721941

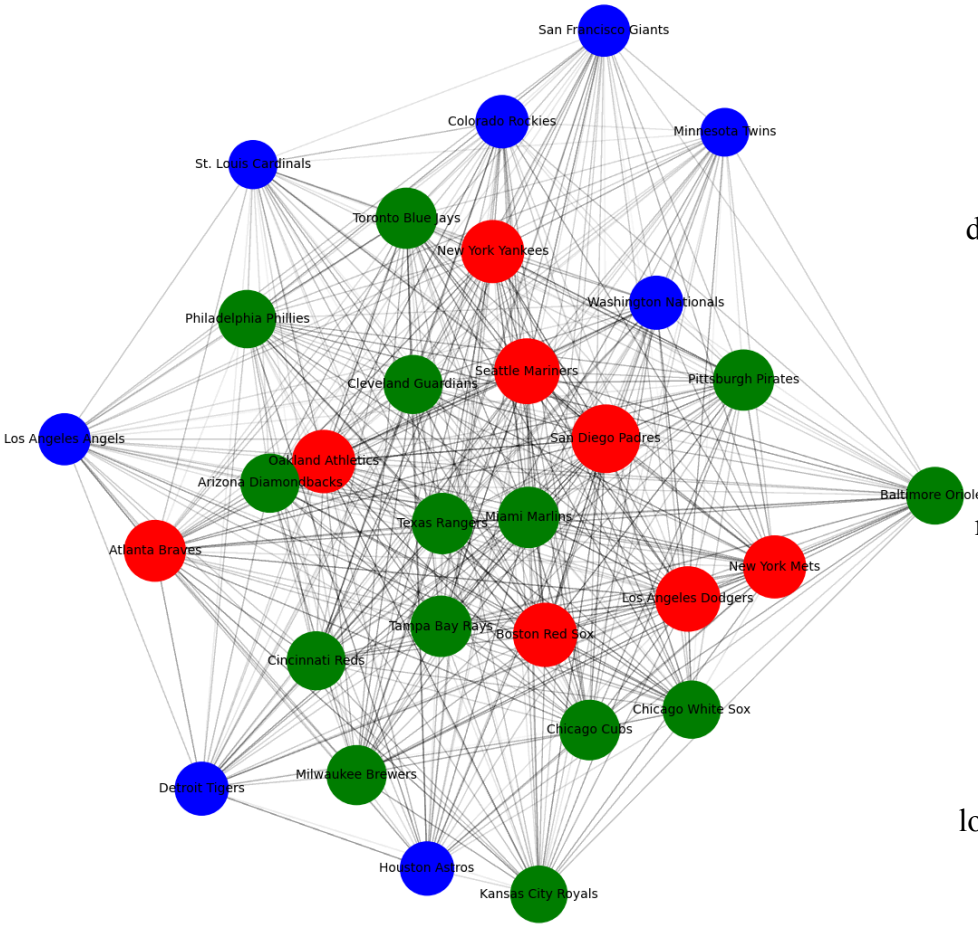
After looking at this data and the lack of correlation, I decided to calculate the average correlation for each team. As it turns out, all centralities contain the same value, besides the degree to which I graphed. Below are also the top and bottom five teams for degree centrality.

	Top in Degree	Bottom in Degree
1	San Diego Padres	Minnesota Twins
2	Seattle Mariners	St. Louis Cardinals
3	Los Angeles Dodgers	Los Angeles Angels
4	Boston Red Soxs	Colorado Rockies
5	Oakland Athletics	Detroit Tigers



correlation 0.45974117161501166

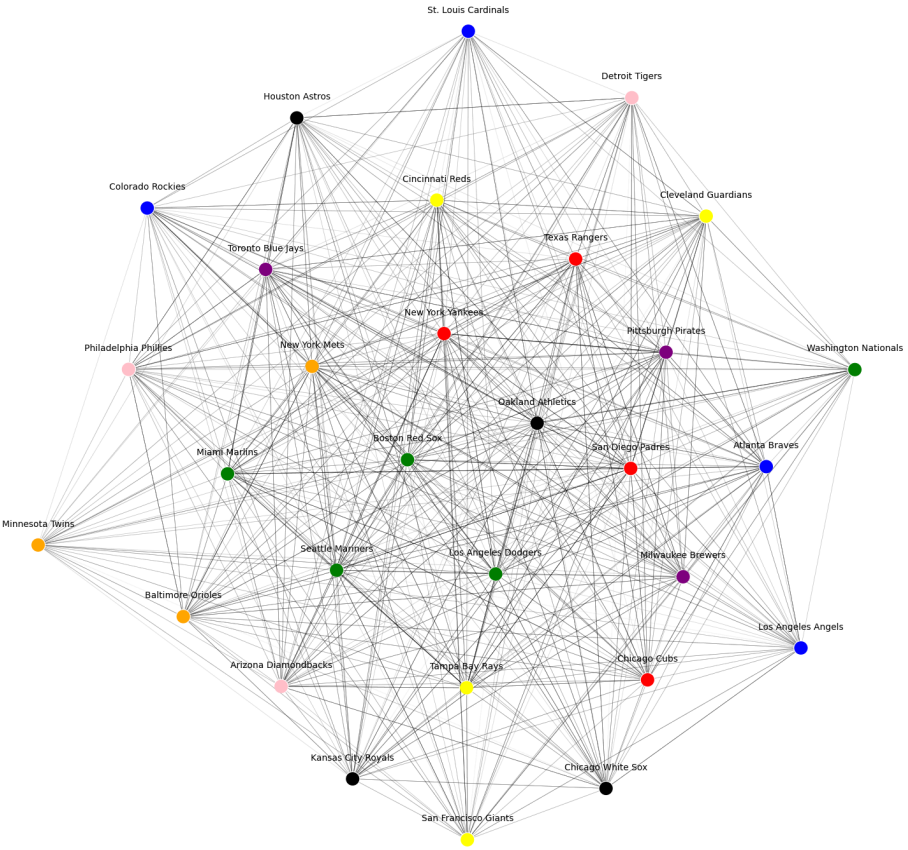
From this data, there is a slight positive correlation, meaning that the more trades the team has as slight correlation to winning more games. After analyzing the graph and looking at the top and bottom teams, the top five have an average winning percentage of 0.535, while the bottom five teams have an average of 0.516. So there is a very slight correlation, around 2%, between having higher degree centrality (trading more) and being more successful.



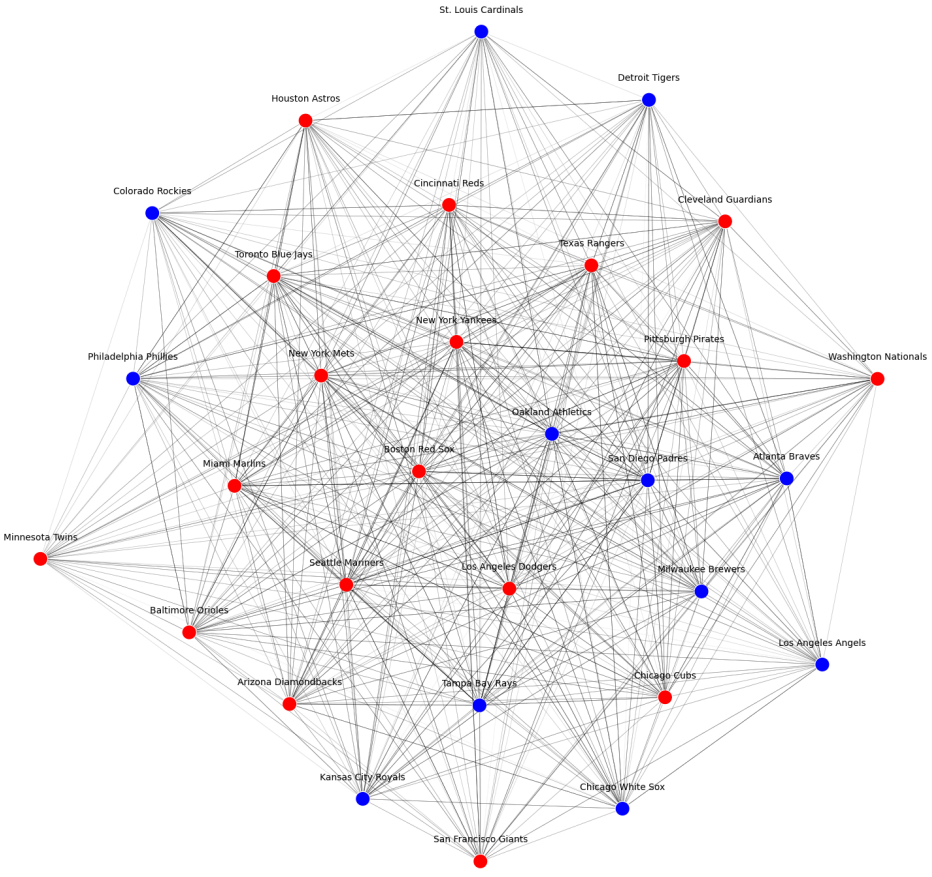
To better visualize this degree centrality, I made this network to demonstrate the degree of centrality of MLB teams over the past 31 seasons being the number of trade connections. Each node is sized accordingly, based on the teams degree centrality. The nodes are then color-coated based on high, medium, and low degree centrality, being red, green, and blue to better distinguish between the teams. With most teams falling within the medium category (green). having two small groups represent a highly connected hub (red) and a loosely connected connect hub blue)

I then used spectral clustering to identify natural clusters in the dataset. I used three clusters in the data set to attempt to find the highest, normal, and lowest trading teams. While this was the plan, my data didn't seem to fit well with spectral clustering. With all the edges and nodes connected, there wasn't much for the program to go on. Two clusters wouldn't show the two leagues, since trading isn't limited to leagues, and 8 clusters wouldn't show divisions, since teams don't often trade within them. Therefore, 3 clusters!

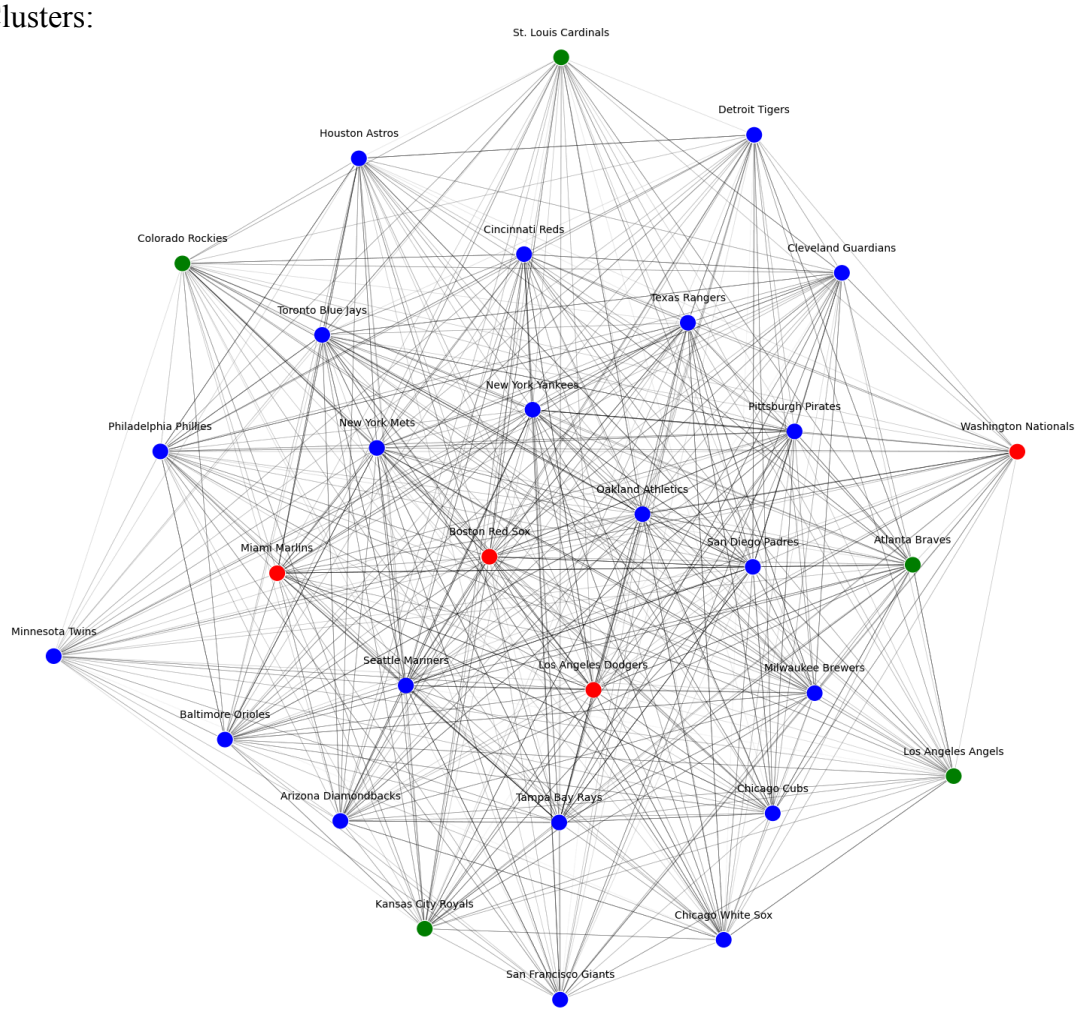
Eight Clusters:



Two Clusters:



Three Clusters:

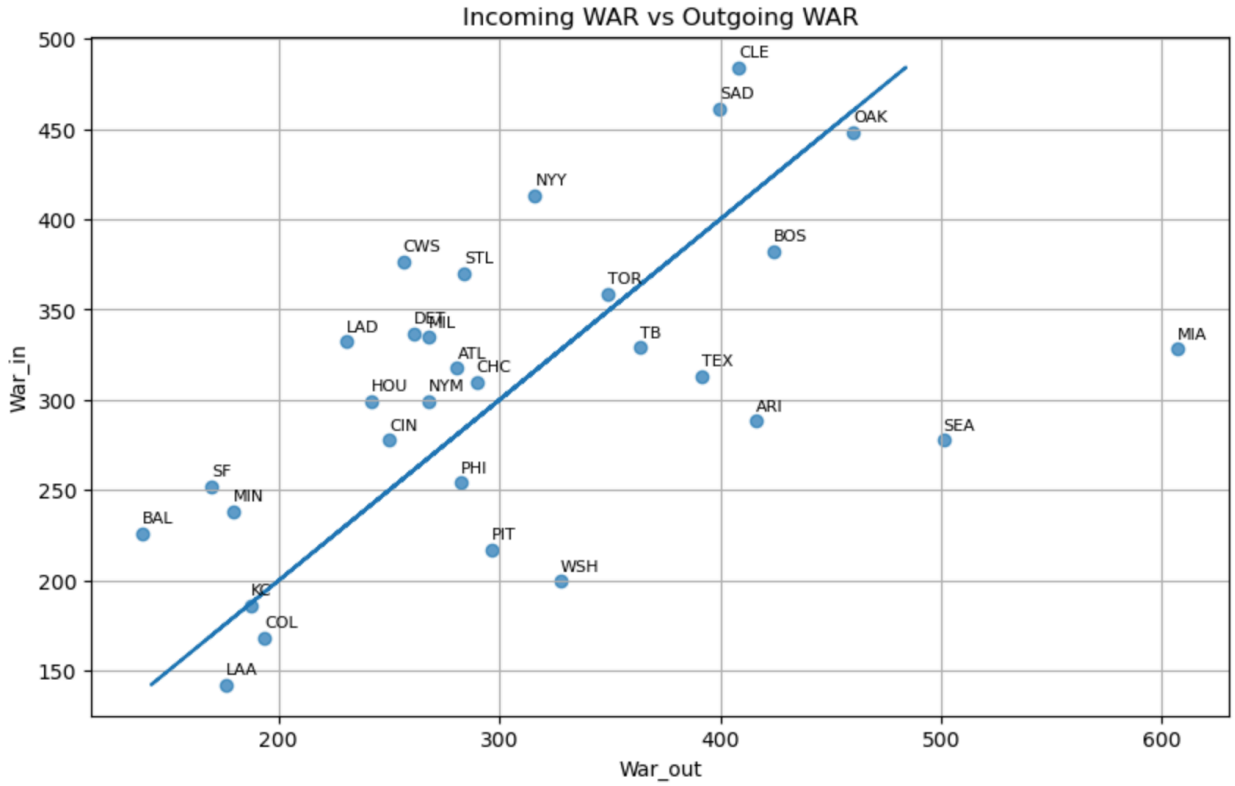


The graph containing three clusters shows that most teams fall into this blue category, most likely indicating normal trading habits. Regarding the red and green groups, I am unsure of the distinction. I looked through the dataset, attempting to find a connection between the teams, but found none. These almost random groups are most likely due to the high number of nodes, resulting in skewed data. So, overall, I do not believe that clustering best fits these datasets.

PART 3:
Applying the Network Model

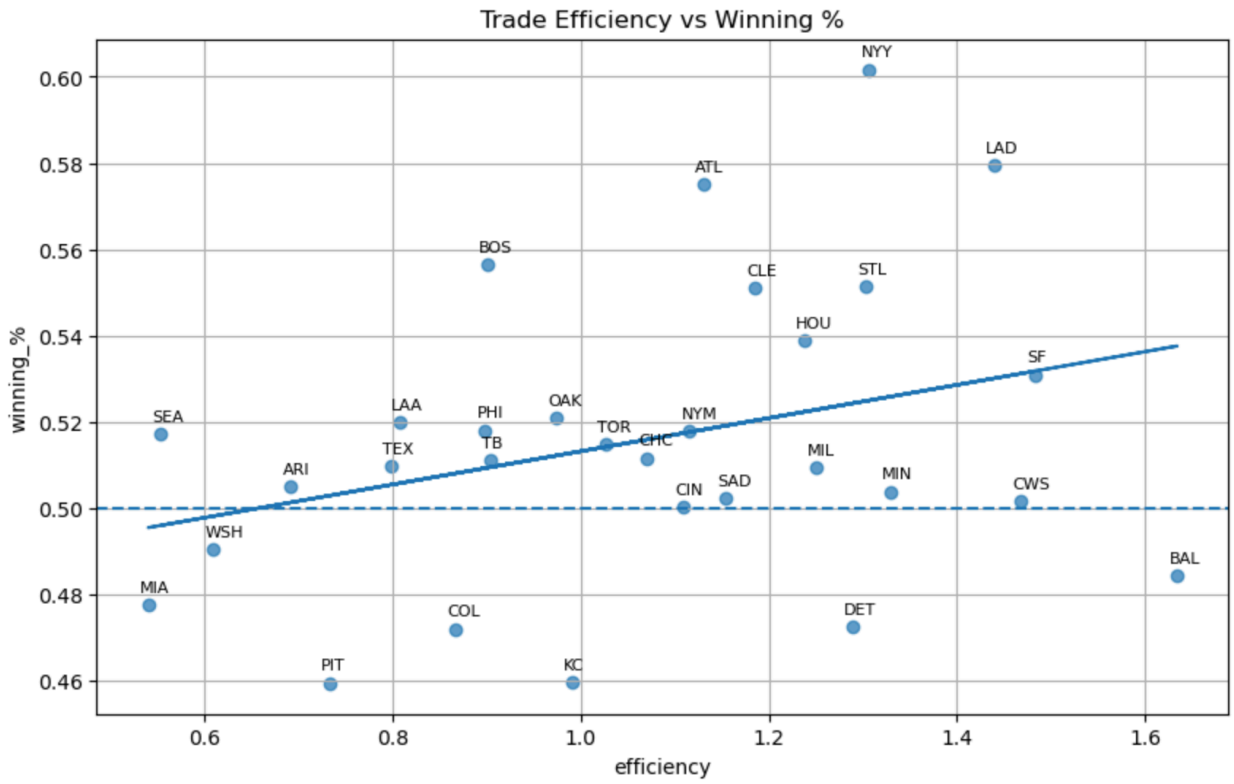
After creating this MLB trade network, I want to analyze it further to see how trading impacts a team's chances of success. Specifically, I will test the idea that more efficient trading teams, measured by winning percentage and WAR, are more successful than less efficient ones, demonstrating that trades can influence a team's trajectory.

To begin this test, I wanted to see which teams were the “successful” over the past 31 seasons in terms of WAR. To do this, I made a scatterplot of the network, organized by incoming WAR vs outgoing WAR. I then added the line y = x to better distinguish success: teams above the line are “successful” at trading, and those below aren't.



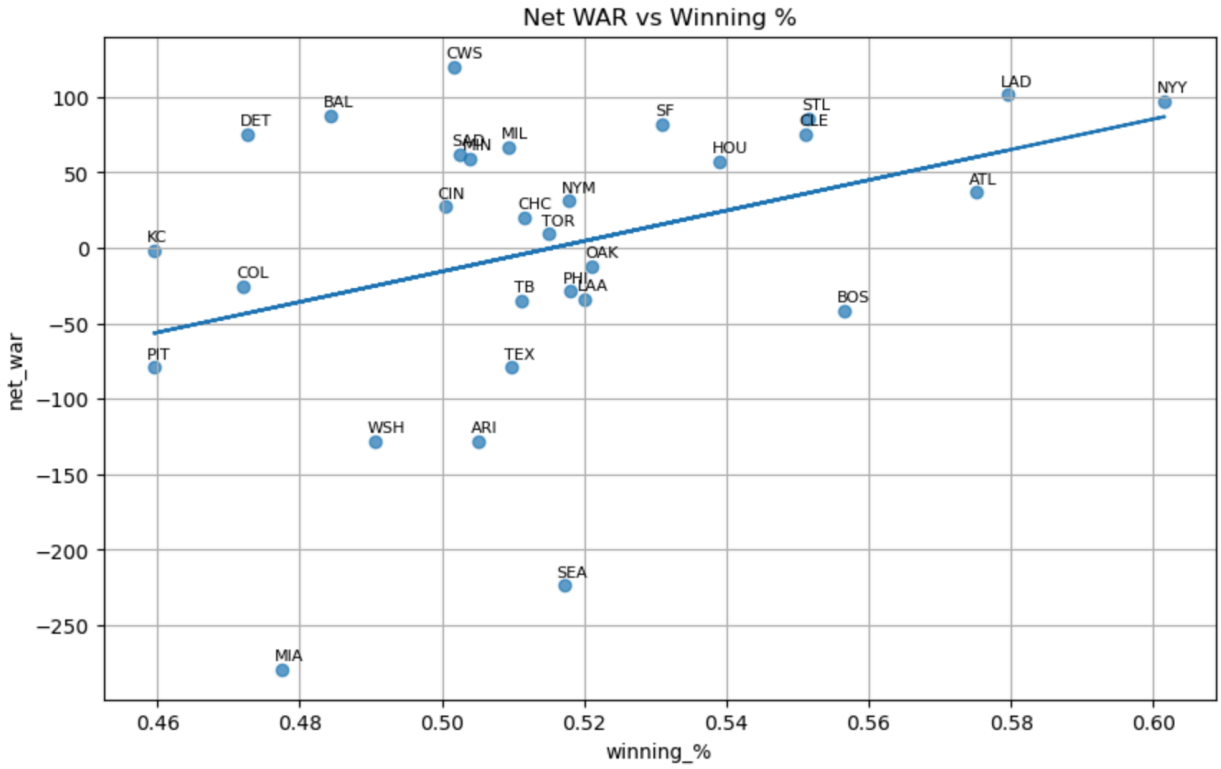
From this graph, it is clear that more teams tend to gain WAR from trades than give it away, with a few extreme outliers driving this overall success. This suggests that while trades often provide a net benefit, the outcomes are uneven across teams. This creates a good baseline for my test to see whether the winning percentage follows the same trend and can predict team success.

For my next graph, I decided to plot trade efficiency vs winning percentage. To do this, I had to find efficiency, which I did by dividing the received WAR by the outgoing WAR. So, efficiency above 1.0 means the teams are more efficient at trading, and below 1.0 means they are not efficient in terms of WAR. I then took each team's average winning percentage over the last 31 seasons and compared it with efficiency to determine whether more efficient teams win more games than less efficient teams. I then added a trendline to better visualize any correlation and a line at 0.500 to show which teams have a winning record (above) and a losing record (below).



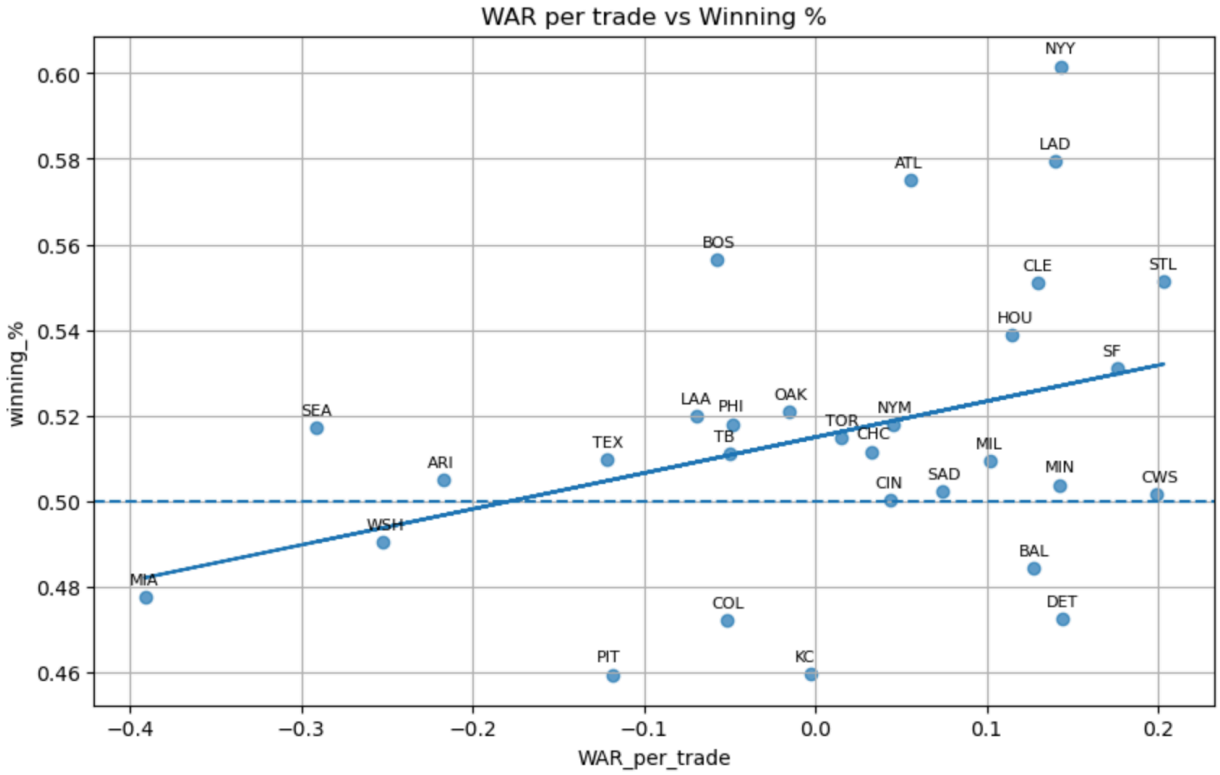
Reading this graph, it shows that there is a slight positive relationship between winning percentage and trade efficiency. The data points, however, are very scattered, with multiple outliers: the Yankees and Dodgers posting extremely high winning percentages and high trade efficiency, while other teams with similar efficiency have much lower winning percentages. This shows that trade efficiency alone is not a strong predictor of team success. Also, if you were to cut the few outliers that dramatically exaggerate the trendline, there doesn't seem to be much of a relationship between trade efficiency and the team's success.

After getting these lackluster results, I wanted to check in general whether trades even matter for winning games. To do this, I decided to graph net WAR vs winning percentage. By doing this, I would be able to see whether teams with more trade WAR were more successful than those without. So to get net WAR, I simply took WAR in and subtracted it from WAR out, then graphed it with the same winning percentage.



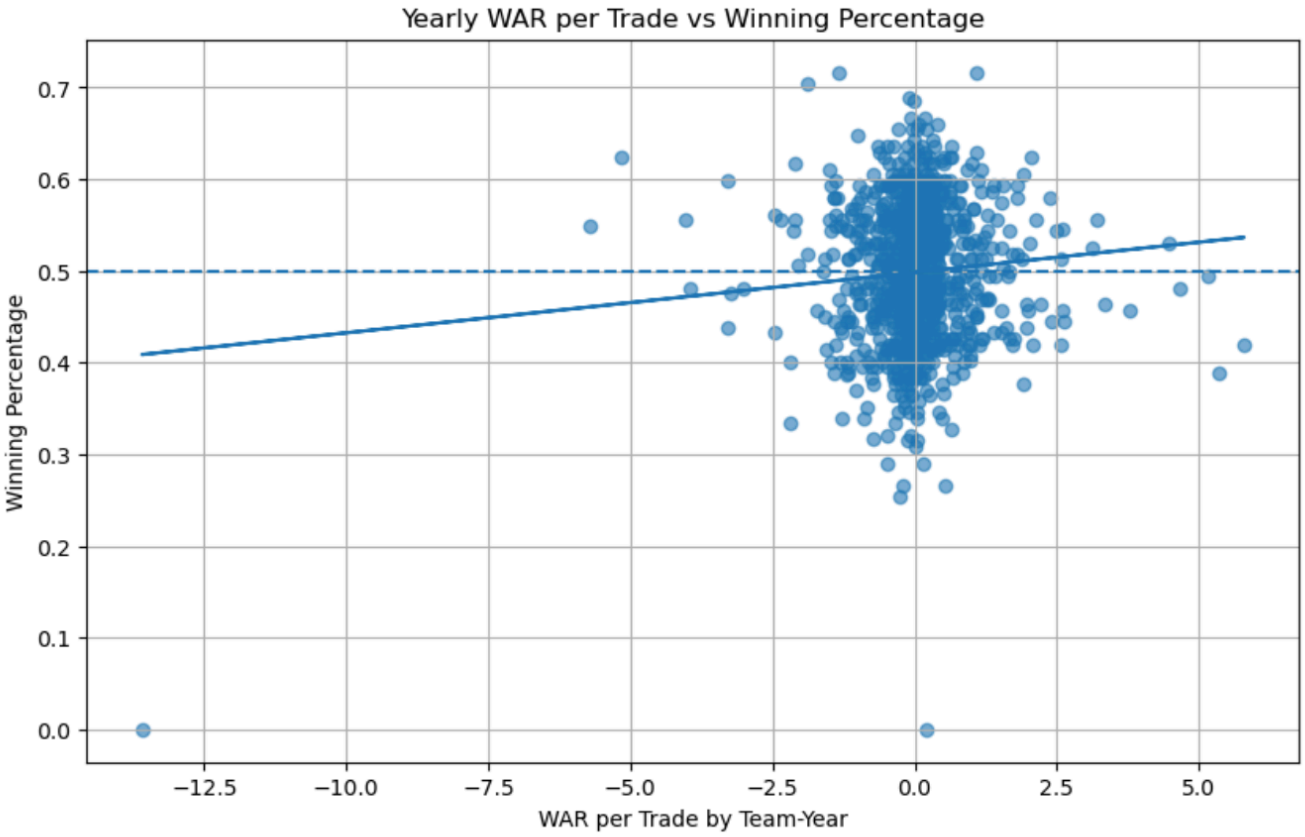
This graph suggests a slight positive relationship between WAR from trades and winning percentage. However, the data is widely dispersed, indicating a weak relationship between net WAR and winning percentage. With multiple teams around 0.52 in winning percentage and a wide range of WAR, and then multiple teams with around 100 WAR and an even wider range of winning percentage. The Yankees (NYY) and Baltimore (BAL) have around a 10% difference in winning percentage despite similar amounts of WAR. This graph is influenced by trade volume, as teams with more trades accumulate larger total WAR values. This introduces bias, making it difficult to isolate the effect of trade quality on team performance. So teams like the Yankees, which have been around forever, compared to the Rays (TB) and Diamondbacks (ARI), who were created in 1998, have less opportunity to trade.

To address the flaws in the previous graph, I decided to plot WAR per trade vs. winning percentage. By graphing WAR per trade, the volume issue would be taken out of the equation, and we could focus purely on trade quality. To find this value, I took net WAR and then divided it by the number of trades that the team had. By normalizing WAR by number of trades, it will isolate trade efficiency and remove the influence of trade volume.



As it turns out, there is once again a weak positive relationship. Similar to the trade efficiency graph, this graph doesn't show a strong correlation between WAR per trade and winning percentage. This is shown well with the White Sox (CWS) and Cardinals (STL): while both have the highest WAR per trade, around 0.2, the Cardinals have been much more successful, boasting a winning percentage 5% higher than the White Sox. Alongside this, the Cardinals have won two World Series and have appeared in four, being a consistent playoff team. While the White Sox have won one World Series, they have never made it past the opening round in the playoffs the rest of the seasons, and they've also been called the “worst team ever” for having the worst winning percentage ever in a season in 2024.

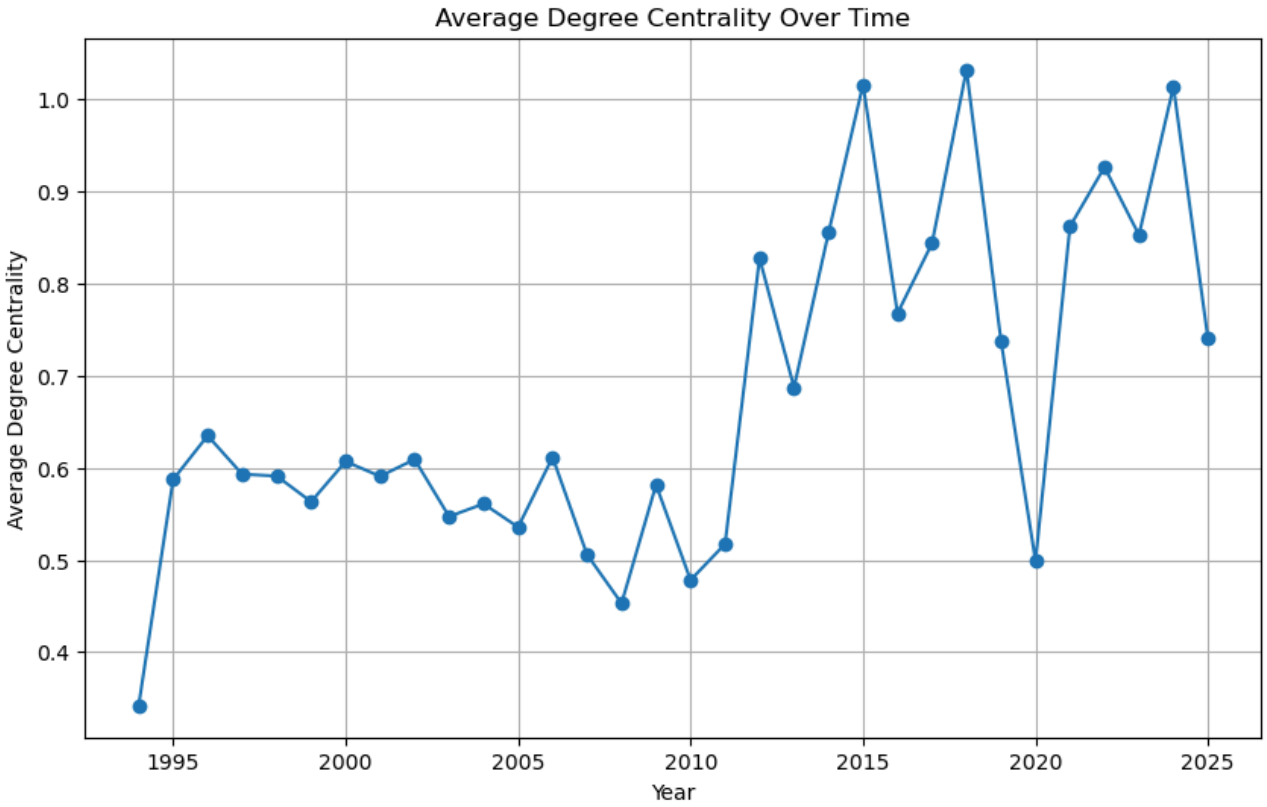
After finding little correlation between WAR per trade and winning percentage, I edited the graph to plot WAR per trade by team each year to see if the averaged data obscured potential findings.




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correlation: 0.08568347698721597
```

This graph shows the relationship between trade efficiency (WAR per trade) and team performance on a yearly basis. While there is a slight positive trend, the data is widely dispersed, and the correlation is very weak at 0.086. This indicates that trade efficiency has little to no predictive power for team success. Teams with similar trade efficiency can have significantly different winning percentages, suggesting that other factors beyond trading play a larger role in determining performance.

I then switched gears to check how degree centrality changed over the course of the past 31 MLB seasons. Degree centrality measures how many trade relationships each team has, so by taking the average, we can see whether teams have become more connected over time and whether that matters.



This graph shows how the connectivity of the MLB trade network evolves over time. The results show that teams have become more connected in recent years, suggesting increased trade activity. However, the fluctuations in recent years also show that this connectivity is not consistent, indicating variability in trading behavior across seasons. As to why the centrality seems to be consistently spiking in recent years, I suspect it is due to shorter term contracts becoming more prominent in the game, alongside the collective bargaining agreements in 1994-1995 and 2011-2012 between players and owners, as well as the pandemic in 2020. That said, more research would be needed to determine the main cause. As to what this means, the network remains relatively stable and highly connected across seasons. This suggests that teams consistently engage in trades at similar levels, which helps explain why centrality measures and WAR do not show strong relationships with team success in the previous analyses.

That being said, it is safe to say that there is not enough evidence to conclude that teams that trade more or are more efficient in trading, in terms of WAR, are more successful than teams that don't trade as much. While getting better players through trades can and has benefited teams, trades can only do so much to improve a team's chances of winning, as the overall team is more important than the individual player added. Factors such as player development, roster construction, and overall team composition play a larger role in determining success than trade activity alone. So if a team is fundamentally flawed and not winning many games, getting a player who gives you 5 WAR per season can't make up for the other 8 players on the field who give you -1 WAR per season.

Discussion Log

- C0
- C1
- C2
- C3
- Office Hours
- Class notes and Slides
- Google Sheets

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“I have a CSV file of all of the MLB trades from 1994 -2025. I want to find the WAR (wins above replacement) each player has provided to the team that traded for them. I want to use bWAR from Baseball Reference. What is the best way to sort through all the player data for all 9000+ data points and find their post-trade war?” response. Gemini, 7 Apr. 2026, Google, gemini.google.com.

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